

Ajay Singh

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Education

Rochester Institute of Technology, B.S. in Software Engineering Dec 2027

- **Relevant Coursework:** Engineering of Software Subsystems, Software Process and Software Management, Embedded Software Engineering, Computer Architecture

Experience

Software Engineering Intern, MigaLabs – Barcelona, Spain Jun 2026 – Aug 2026

- Developed and deployed a full-stack social media analytics platform using **React**, **Express.js**, **MongoDB Atlas**, and **Docker**.
- Implemented scheduled data collection workflows with **cron jobs** to automatically retrieve and update analytics from external **REST APIs**.
- Engineered automated data ingestion pipelines to collect, store, and analyze social media engagement metrics.
- Designed trend-analysis algorithms to identify emerging topics and audience interests for data-driven content strategy.
- Built interactive dashboards with **Recharts** to transform complex social media datasets into actionable business insights.

Projects

Bits N' Bytes (Ongoing) GitHub

- Developed a self-service retail kiosk using **OpenCV** and **YOLO** object detection with **Python** and **C++**.
- Developed AI functionality in **Python** as part of a 3-person software team and collaborated with multiple software teams to integrate the system.
- Engineered a touchscreen UI using **Qt Creator** and **Raspberry Pi** hardware.
- Developed cabinet interaction protocols using **ESP32** microcontrollers and **Python MQTT** to automate cabinet operations.

LJHS – Homeless Shelter Website

- Developed a full-stack donation platform using **Angular** and **Spring Boot** as part of a 4-person **Agile/Scrum** development team.
- Participated in sprint planning and daily stand-ups throughout a multi-month software development cycle.
- Designed and implemented **RESTful APIs** using Spring Boot to support application functionality and backend services.

NoteAI GitHub

- Developed a **Flask** web application that converts MP3 audio to text using **FFmpeg** and the **Microsoft Azure API**.
- Implemented multiple text-processing outputs including **abstractive summarization**, **extractive summarization**, and **full transcription**.

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Angular, Flask, Spring Boot

Technologies: Docker, Git, MongoDB, PostgreSQL, REST APIs, MQTT, OpenCV, YOLO, FFmpeg, Microsoft Azure